60 DAYS | QUALITY ENGINEERING LAB

LABORATORY EXPERIMENT REPORT

EXPERIMENT NO. 02 — PROCESS FAILURE MODE & EFFECTS ANALYSIS (PFMEA)

Systematic Identification and Evaluation of Potential Manufacturing Process Failures in CNC Machining of a Precision Automotive Shaft

> **SIMULATED INDUSTRIAL TRAINING DATA NOTICE:**
> This laboratory experiment report has been compiled as part of an independent Quality Engineering professional curriculum. All manufacturing parameters, failure frequencies, Severity/Occurrence/Detection ratings, measurement tolerances, and financial metrics presented herein are derived from realistic simulated industrial engineering models. They do not represent proprietary data from any specific automotive OEM or Tier-1 supplier.

Curriculum Track:	60 Days Quality Engineering Lab
Engineering Discipline:	Manufacturing Quality Engineering & Process Risk Management
Target Product:	EN19 Alloy Steel Precision Transmission Input Shaft
Core Methodology:	AIAG & VDA FMEA Handbook (1st Edition) / Traditional RPN & Action Priority (AP)
Analytical Tools:	PFMEA Worksheet, 5-Why Analysis, 6M Fishbone Diagram, Pareto Risk Chart, Control Plan, CAPA
Author / Role:	Quality Engineer
Date of Execution:	November 2026

SECTION 1: EXPERIMENT OVERVIEW

1.1 Aim of the Experiment

The primary aim of this experiment is to systematically analyze the multi-stage CNC manufacturing process of a precision automotive transmission shaft using Process Failure Mode and Effects Analysis (PFMEA). The objective is to identify potential process failure modes, quantitatively evaluate initial risks using Severity (S), Occurrence (O), and Detection (D) ratings, determine traditional Risk Priority Numbers (RPN) and AIAG-VDA Action Priority (AP) levels, formulate robust corrective and preventive action plans (CAPA), and verify residual risk reduction.

1.2 Detailed Objectives

- Deconstruct the 9-stage manufacturing flow of a precision alloy steel shaft into discrete process activities, functions, and key characteristics.
- Systematically identify at least 10 realistic process failure modes across turning, facing, drilling, grinding, and inspection steps.
- Establish evidence-based 1–10 rating scales for Severity, Occurrence, and Detection aligned with industrial quality standards.
- Calculate initial RPN values and evaluate Action Priority (AP) for each failure mode to identify high-risk manufacturing operational bottlenecks.
- Perform root cause analysis using 5-Why iteration and 6M Fishbone (Ishikawa) diagrams for critical defects like shaft diameter variation and tool chipping.
- Formulate targeted preventive controls, Poka-Yoke mistakeproofing mechanisms, and advanced detection protocols.
- Synthesize a comprehensive Control Plan, CAPA report, and shop-floor Inspection Checklist to govern post-improvement production.

1.3 Problem Statement

High-precision automotive transmission components operate under severe mechanical fatigue, torsional stress, and rotational speeds exceeding 6,000 RPM. Dimensional deviations as small as ±0.015 mm or surface roughness exceeding Ra 0.8 µm in critical bearing journals lead to catastrophic transmission noise, vibration, premature wear, oil seal leakage, and field failure. Uncontrolled process variability during CNC turning and grinding operations results in high scrap rates, unexpected tool failures, and customer line stops. A proactive, structured risk mitigation framework is required prior to mass production launch.

1.4 Scope of the Experiment

The scope encompasses the full physical manufacturing sequence of an EN19 forged alloy steel shaft, starting from Raw Material Receiving & Verification (Stage 1) through CNC Turning, Length Machining, Drilling, Precision Grinding, Deburring, and Final Quality Verification (Stage 9). Component design architecture (DFMEA) is treated as an input boundary condition; the analysis focuses strictly on process-induced failure modes, tooling wear, clamping errors, machine thermal drift, and operator handling factors.

1.5 Industrial Background & Importance of PFMEA

In modern lean automotive manufacturing, quality cannot be inspected into a product; it must be designed into the manufacturing process. PFMEA serves as a analytical risk management tool that bridges process design and production control. By evaluating potential failure modes prior to tooling sign-off, engineering teams prevent costly warranty recalls, minimize internal scrap and rework costs, fulfill IATF 16949 quality system requirements, and ensure zero-defect product launches.

1.6 Expected Learning Outcomes & Engineering Relevance

Practitioners gain hands-on expertise in executing end-to-end PFMEA workflows, differentiating between prevention and detection mechanisms, applying mathematical risk scoring, constructing Pareto prioritization charts, and translating analytical risk findings into executable shop-floor Control Plans and CAPA documentation.

SECTION 2: INDUSTRIAL APPLICATION

2.1 Technical Profile of a Precision Automotive Shaft

A precision shaft is a cylindrical mechanical component designed to transmit torque, rotational power, and angular velocity between powertrain sub-assemblies. In automotive transmissions and drivetrain systems, input shafts support rotating gears, transmit engine power to the gearbox, and interface directly with bearings and oil seals.

Technical Attribute	Specification / Requirement	Engineering Significance
Material Grade	EN19 (AISI 4140) Alloy Steel (Induction Hardened)	Provides high tensile strength, core toughness, and wear resistance.
Critical Journal Diameter	Ø35.000 mm (+0.000 / -0.012 mm) [IT6 Precision]	Ensures precise interference fit with high-speed rolling element bearings.
Total Radial Runout	≤ 0.010 mm TIR relative to datum centerlines	Prevents rotational imbalance, high-frequency vibration, and bearing fatigue.
Surface Roughness (Ra)	Ra ≤ 0.4 µm (Bearing Journal), Ra ≤ 1.6 µm (Spline)	Minimizes dynamic friction, heat generation, and lip seal degradation.
Overall Finished Length	250.00 mm ± 0.05 mm	Guarantees correct axial stack-up clearance within transmission housing.

2.2 Impact of Machining Errors on Vehicle Performance

Machining errors propagate directly into functional vehicle failures. Oversized bearing journals cause bearing seizure during press-fitting, while undersized journals result in shaft slippage and dynamic fretting corrosion. Excessive radial runout creates severe NVH (Noise, Vibration, Harshness) issues, accelerating gear tooth pitting and seal breakdown. Poor surface finish causes rapid friction wear on rotary lip seals, inducing transmission fluid leaks and eventual gearbox destruction.

SECTION 3: COMPLETE MANUFACTURING PROCESS FLOW

The production sequence comprises 9 distinct manufacturing stages. Each step is engineered with specific parameters and quality gates to satisfy engineering drawing specifications.

Seq	Process Stage	Core Activity & Purpose	Key Process Parameters	Quality Requirements
1	Raw Material Receipt	Verify raw forged EN19 round bar stock chemistry, hardiness, and dimensions.	Spectrometry chemical composition, Brinell hardness (HBW), bar stock straightness.	Material Grade: EN19; Hardness: 180-220 HBW; Zero surface seams/ cracks.
2	Raw Material Sawing	Cut round bars into individual billet lengths for CNC turning.	Band saw blade speed (45 m/min), feed rate (30 mm/min), coolant concentration (6-8%).	Billet Length: 255.0 ± 0.5 mm; Squareness: ≤ 0.2 mm.
3	CNC Rough Turning	Machine main outer steps and center locations from cut billet.	Cutting speed Vc = 180 m/min, Feed f = 0.25 mm/rev, Depth of cut ap = 2.0 mm.	Rough Diameter: Ø35.8 ± 0.1 mm; Concentricity: ≤ 0.05 mm.
4	CNC Finish Turning	Finish turn bearing journals, chamfers, and snap ring grooves.	Vc = 240 m/min, f = 0.10 mm/rev, ap = 0.4 mm, Tool insert wear monitoring.	Finish Diameter: Ø35.05 ± 0.02 mm (Pre-grind); Surface Finish: Ra ≤ 1.6 µm.
5	Drilling / Center Boring	Drill axial oil lubrication hole and machine precision center holes.	Drill RPM = 2200, Feed rate = 0.08 mm/ rev, High-pressure internal coolant (20 bar).	Hole Diameter: Ø6.0 ± 0.1 mm; Center Angle: 60° ± 0.25°.
6	Precision Cylindrical Grinding	Final grinding of bearing journals to exact drawing tolerance and finish.	Wheel speed = 45 m/s, Workhead RPM = 150, Infeed rate = 0.002 mm/pass, Dressing cycle.	Journal Diameter: Ø35.000 (+0/-0.012 mm); Ra ≤ 0.4 µm; Runout ≤ 0.010 mm.

Seq	Process Stage	Core Activity & Purpose	Key Process Parameters	Quality Requirements
7	Deburring & Washing	Remove machining burrs and ultrasonic wash to remove chips and grinding swarf.	Deburring brush RPM = 1200, Wash temp = 60°C, Ultrasonic power = 1.2 kW, Drying air.	Zero burrs under 10x magnification; Millipore cleanliness ≤ 2.0 mg/part.
8	In-Process Inspection	Perform 100% air gage diameter checks and multi-gage runout inspection.	Air gage calibration pressure, Pneumatic probe set point, Master ring calibration.	Diameter within IT6 tolerance; Runout ≤ 0.010 mm; Cpk ≥ 1.67.
9	Final Quality Verification & Packing	Final QA dock audit, rust preventative coating, and VCI protective packaging.	Audit sampling size (AQL 0.65 Level II), Dip tank oil concentration, VCI bag sealing.	100% compliance to drawing; Zero surface corrosion; Approved QA stamp.

SECTION 4: PFMEA FUNDAMENTALS

4.1 Core Definitions & Conceptual Framework

Process Failure Mode and Effects Analysis (PFMEA): A structured analytical risk-management methodology utilized by engineering teams to identify, evaluate, and mitigate potential process-related failures before production execution.

Parameter	Engineering Definition	Manufacturing Context / Example
Failure Mode	The specific physical manner in which a process step fails to meet its functional requirement.	Finish turning yields an outer journal diameter of Ø35.045 mm (oversized relative to upper specification limit of Ø35.020 mm).
Failure Effect	The direct impact or consequence of the failure mode on downstream operations, the assembly plant, or the ultimate end-user.	Transmission assembly plant line stop due to inability to press-fit bearing onto oversized journal; potential casing cracking.
Failure Cause	The specific thermal, mechanical, physical, or human mechanism that induces the failure mode.	Gradual flank wear of the carbide turning insert exceeding 0.3 mm Vb threshold, causing insert edge recession.
Severity (S)	Numerical rating (1–10) estimating the seriousness of the effect on the customer/process.	Rating = 7: Customer line stop > 1 hour, scrap component, potential mechanical fit failure.
Occurrence (O)	Numerical rating (1–10) estimating the likelihood that the specific failure cause will occur.	Rating = 4: Occasional failure mode, estimated rate ~ 1 in 1,000 pieces based on historical SPC data.
Detection (D)	Numerical rating (1–10) estimating the capability of existing control methods to detect the cause/mode.	Rating = 8: Manual sample check using vernier caliper at end of shift (high probability of passing non-conforming parts).
RPN	Traditional Risk Priority Number, calculated as $RPN = S \times O \times D$ (Scale 1 to 1000).	$RPN = 7 \times 4 \times 8 = 224$. High numerical risk requiring prioritised corrective action.
Action Priority (AP)	AIAG-VDA logic-based prioritization methodology (High, Medium, Low) replacing simple RPN.	Evaluates S, O, D combination systematically, giving absolute priority to High Severity items.

4.2 Key Engineering Distinctions

- **Prevention Controls vs. Detection Controls:** Prevention controls eliminate or reduce the likelihood of the failure cause occurring (e.g., tool life auto-counter that locks machine after 200 cuts). Detection controls identify the presence of a defect or cause after it has occurred (e.g., downstream 100% air gaging station).
- **Failure Mode vs. Failure Cause:** The failure mode is the non-conforming condition of the workpiece (e.g., Excessive Runout), whereas the failure cause is the root process mechanism (e.g., Swarf trapped between fixture locator and shaft center).
- **Initial Risk vs. Residual Risk:** Initial risk represents the baseline process condition prior to intervention. Residual risk represents the calculated risk remaining after implementing verified CAPA actions.

SECTION 5: SEVERITY, OCCURRENCE AND DETECTION RATING SYSTEM

Disclaimer: The following criteria represent custom, illustrative training rating scales developed for this engineering laboratory experiment, inspired by AIAG-VDA risk principles. They do not replace official published AIAG-VDA standard tables.

Table 5.1: Severity (S) Rating Criteria (1–10 Scale)

Rating	Impact Level	Effect on Product / Assembly Customer	Effect on Manufacturing / Plant Operation
10	Hazardous (No Warning)	Affects safe vehicle operation or compliance with government regulations without prior warning.	Severe hazard to operator safety (e.g., projectile part, major machine explosion).
9	Hazardous (With Warning)	Affects safe vehicle operation or compliance with government regulations with prior warning.	Operator safety risk with warning devices activated. Major regulatory non-compliance.
8	Very High	Vehicle rendered totally inoperable; loss of primary function (transmission failure).	100% of production run scrapped. Major assembly line stop exceeding 4 hours.
7	High	Vehicle operable but at reduced performance level. Severe customer dissatisfaction.	Product must be sorted and 100% reworked. Line stop between 1 to 4 hours.
6	Moderate	Vehicle operable; comfort/convenience items inoperable. Customer experiences discomfort.	Portion of production run scrapped (10-20%). Rework required off-line.
5	Low	Vehicle operable; minor annoyance (e.g., slight transmission gear hum/noise).	Product must be sorted; partial scrap (< 10%). Minor inline rework.
4	Very Low	Fit & finish or appearance item does not conform. Noticed by most customers (>75%).	Minor defect; product accepted with minor inline re-machining.
3	Minor	Fit & finish item non-conforming. Noticed by discriminating customers (50%).	Minor process inconvenience; slight machine operator adjustment needed.
2	Very Minor	Fit & finish item non-conforming. Noticed by very few customers (<15%).	Minor process inconvenience; zero scrap or rework generated.
1	None	No discernible effect on product performance, assembly, or end customer.	No operational impact. Completely imperceptible process deviation.

Table 5.2: Occurrence (O) Rating Criteria (1–10 Scale)

Rating	Likelihood of Failure	Estimated Failure Rate (PPM / Frequency)	Process Prevention Capability
10	Extremely High	≥ 100,000 PPM (≥ 1 in 10 parts)	No prevention controls. Unstable process, highly variable.
9	Very High	50,000 PPM (1 in 20 parts)	Weak prevention controls; frequent parameter shifts.
8	High	20,000 PPM (1 in 50 parts)	Manual process controls; reliance on operator vigilance.
7	Moderately High	10,000 PPM (1 in 100 parts)	Basic SOPs in place; periodic manual tool offsets.
6	Moderate	2,000 PPM (1 in 500 parts)	Standard CNC control; tool life estimated by experience.
5	Low-Moderate	500 PPM (1 in 2,000 parts)	Automated process control; preventive maintenance program.
4	Low	100 PPM (1 in 10,000 parts)	Poka-Yoke prevention design; automated tool life counters.
3	Very Low	10 PPM (1 in 100,000 parts)	Robust closed-loop CNC feedback; high process capability (Cpk ≥ 1.67).
2	Remote	1 PPM (1 in 1,000,000 parts)	Error-proofed fixture design; fully automated loading & sensing.
1	Extremely Remote	< 0.1 PPM (Failure physically impossible)	Absolute physical prevention by process design (Poka-Yoke).

Table 5.3: Detection (D) Rating Criteria (1–10 Scale)

Rating	Detection Opportunity	Detection Control Method	Probability of Detection
10	Absolute Uncertainty	No detection control in place; defect cannot be detected.	0% — Defect passed directly to customer.
9	Very Remote	Visual audit by operator on sample basis (un-standardized).	Very low detection probability (< 20%).
8	Remote	Manual measurement using hand tools (calipers/micrometers) at shift end.	Low detection capability; subject to human error.
7	Very Low	Manual attribute gaging (Go/No-Go) on periodic sample basis.	Moderate-low detection; post-process offline.
6	Low	Manual variable gaging at station on 100% of parts offline.	Moderate detection capability.
5	Moderate	Automated offline gaging station with SPC variable charting.	High probability of detecting process shifts.
4	Moderately High	Automated in-line gaging station checking 100% of parts post-process.	Very high detection; stops machine upon defect.
3	High	In-process automated probe sensing prior to part ejection.	Near 100% detection; prevents part exiting machine.
2	Very High	Automated closed-loop inline sensing that automatically rejects non-conforming part.	100% immediate detection and isolation.
1	Almost Certain	Error-proofing (Poka-Yoke) prevents processing of non-conforming part.	Defect cannot physically proceed downstream.

SECTION 6: PFMEA METHODOLOGY — STEP BY STEP

Executing a robust PFMEA requires adherence to a 15-step structured engineering methodology, systematically applied to the CNC shaft production line:

1. **Define Process & Scope:** Establish boundaries covering Stages 1 to 9 of the EN19 shaft line.
2. **Identify Process Functions:** Map required outputs for each operation (e.g., Stage 6: Precision Grind bearing journal to Ø35.000 +0/-0.012 mm).
3. **Identify Potential Failure Modes:** Brainstorm non-conformance modes (e.g., Journal diameter undersized).
4. **Identify Failure Effects:** Trace consequences to downstream assembly and end-user vehicle operations.
5. **Identify Failure Causes:** Determine root physical mechanisms (e.g., Thermal expansion of grinding spindle).
6. **Identify Existing Prevention Controls:** Document active prevention safeguards (e.g., Coolant temperature chiller maintained at 20°C ± 1°C).
7. **Identify Existing Detection Controls:** Document active measurement methods (e.g., Post-grind 100% air gage check).
8. **Assign Initial S, O, D Ratings:** Utilize standardized 1–10 criteria tables to assign objective numerical scores.
9. **Calculate Initial RPN:** Compute $RPN = S \times O \times D$ for baseline prioritization.
10. **Evaluate Action Priority (AP):** Apply AIAG-VDA logic matrices to classify risks into High, Medium, or Low categories.
11. **Prioritize Significant Risks:** Focus engineering resources on High AP items and RPN values exceeding 150.
12. **Recommend Improvement Actions:** Formulate engineering controls (e.g., In-process touch trigger probe implementation).
13. **Assign Responsibility & Dates:** Designate specific owners and target completion schedules.
14. **Reassess Risk Post-Implementation:** Execute actions, re-evaluate S, O, and D ratings, and recalculate residual RPN.
15. **Update Control Plan & Standard Work:** Embed verified controls into shop-floor Control Plans and SOP documentation.

SECTION 7: COMPLETE INDUSTRIAL PFMEA WORKSHEET

The following worksheet summarizes the complete Process Failure Mode and Effects Analysis for 10 realistic failure modes in the CNC shaft manufacturing process.

FM ID	Process Step & Function	Potential Failure Mode	Potential Failure Effect	S	Potential Failure Cause	O	Current Prevention Controls	Current Detection Controls	D	Init RPN	AP	Recommended Action	Rev S	Rev O	Rev D	Rev RPN	Rev AP
FM-01	Stage 4: CNC Finish Turning (Ø35.05mm Journal)	Incorrect Outer Diameter (Oversized Ø35.08 mm)	Grinding wheel loading in Stage 6; potential scrap or excessive grinding cycle time.	7	Insert flank wear exceeding 0.3 mm; failure to update tool offset.	4	Manual tool counter setting by operator experience.	Manual micrometer check 1 per 25 parts.	8	**224**	HIGH	Install RFID auto tool-life counter & in-process post-turn optical probe.	7	2	3	**42**	LOW
FM-02	Stage 6: Cylindrical Grinding (Bearing Journal)	Excessive Shaft Runout (> 0.025 mm TIR)	High transmission NVH; premature bearing failure; customer rejection.	8	Swarf trapped between tailstock center point and center hole.	5	Standard coolant flush nozzle; manual wiping.	Offline dial indicator check 1 per 10 parts.	6	**240**	HIGH	Implement high-pressure air blast center-hole cleaning Poka-Yoke before clamping.	8	2	3	**48**	LOW
FM-03	Stage 4: CNC Finish Turning (Facing to Length)	Incorrect Shaft Overall Length (250.25 mm)	Axial interference during transmission casing assembly; casing cracking.	6	Billet axial locating stop worn or shifted due to clamping impact.	3	Preventive maintenance visual check monthly.	Vernier height gage check 1 per shift.	5	**90**	MED	Replace mechanical stop with hardened carbide stop & automated proximity sensor.	6	2	2	**24**	LOW
FM-04	Stage 6: Cylindrical Grinding (Surface Finishing)	Poor Surface Finish (Ra 0.9 µm vs spec Ra ≤ 0.4 µm)	Rapid oil seal wear; transmission fluid leakage; field warranty claim.	5	Grinding wheel dulling/ glazing due to missed dressing interval.	4	Manual dressing frequency log sheet.	Surface roughness tester check 1 per 50 parts.	8	**160**	HIGH	Automate diamond dressing cycle after every 15 parts via CNC macro code.	5	2	3	**30**	LOW
FM-05	Stage 3: CNC Rough Turning (Heavy Cutting)	Tool Breakage / Chipping during Turning	Severe part gouging; tool holder destruction; spindle shock load.	8	Excessive depth of cut combined with hard inclusion in raw forging.	6	Raw material hardness certificate verification.	Operator visual observation & noise detection.	6	**288**	HIGH	Integrate CNC acoustic emission & spindle power monitoring auto-shutoff system.	8	2	3	**48**	LOW
FM-06	Stage 3: CNC Rough Turning (Diameter Machining)	Chatter Marks on Shaft Outer Surface	Subsequent grinding unable to clean up chatter pattern; aesthetic scrap.	5	Insufficient tailstock hydraulic pressure; loose clamping force.	4	Hydraulic pressure gage manual check at shift start.	Visual inspection post-turning 100%.	6	**120**	MED	Add electronic pressure switch interlocked with CNC cycle start circuit.	5	2	24	**24**	LOW
FM-07	Stage 7: Deburring & Washing (Chamfers)	Burr Formation at Chamfer Edges	Operator hand injury; assembly line jamming; contamination in valves.	4	Worn deburring brush bristles; insufficient contact pressure.	3	Scheduled brush replacement every 2 weeks.	Tactile finger-glove check 1 per 20 parts.	4	**48**	LOW	Standardize brush height setup fixture & implement automated rotary head.	4	2	2	**16**	LOW
FM-08	Stage 1: Raw Material Receipt (Forgings)	Incorrect Material Hardness (140 HBW vs 200 HBW)	Component torsional strength failure under vehicle operating load; catastrophic drive failure.	9	Supplier material mix-up at mill; incorrect heat treatment batch.	4	Supplier Certificate of Analysis (CoA) review per heat lot.	Sample benchtop hardness test 3 bars per lot.	7	**252**	HIGH	Implement 100% inline eddy current hardness sorting station at raw material unloading.	9	2	3	**54**	LOW
FM-09	Stage 6: Cylindrical Grinding (Fine Journal)	Thermal Distortion of Shaft (Out of Roundness)	Tapered journal profile; localized bearing overload; premature wear.	7	Coolant flow restriction or temperature spike in grinding zone (> 30°C).	4	Routine coolant tank level check daily.	Offline roundness measuring instrument (CMM) 1 per shift.	5	**140**	MED	Install coolant inline digital flow sensor & active chiller unit maintained at 20°C ±1°C.	7	2	2	**28**	LOW

FM ID	Process Step & Function	Potential Failure Mode	Potential Failure Effect	S	Potential Failure Cause	O	Current Prevention Controls	Current Detection Controls	D	Init RPN	AP	Recommended Action	Rev S	Rev O	Rev D	Rev RPN	Rev AP
FM-10	Stage 8: In-Process Inspection	Inadequate Inspection Data / False Acceptance	Non-conforming shafts shipped to customer; customer assembly defect spill.	6	Air gage master setting ring worn out; uncalibrated measurement system.	4	Calibration schedule sticker on instrument.	Periodic Quality Auditor supervisory check.	6	**144**	MED	Implement automated master calibration prompt requiring operator zeroing every 4 hours.	6	2	2	**24**	LOW

SECTION 8: RPN CALCULATIONS

8.1 Mathematical Formula

The traditional Risk Priority Number (RPN) is calculated using the equation:

> $$RPN = Severity\ (S) \times Occurrence\ (O) \times Detection\ (D)$$

Where S, O, and D are discrete integers rated on a 1 to 10 scale, yielding RPN values ranging from 1 to 1,000.

8.2 Worked Sample Calculations

- **Calculation 1 — FM-05 (Tool Breakage during Turning):**
  Selected Ratings: $S = 8$, $O = 6$, $D = 6$.
  Substitution: $RPN = 8 \times 6 \times 6 = 288$.
  Engineering Interpretation: Highest baseline risk in the plant. High occurrence due to forging inclusions combined with moderate detection capability requires immediate automated power monitoring controls.
- **Calculation 2 — FM-08 (Incorrect Raw Material Hardness):**
  Selected Ratings: $S = 9$, $O = 4$, $D = 7$.
  Substitution: $RPN = 9 \times 4 \times 7 = 252$.
  Engineering Interpretation: Critical safety/regulatory risk. High Severity (9) indicates potential catastrophic drivetrain failure; sample bench testing provides insufficient detection.
- **Calculation 3 — FM-02 (Excessive Shaft Runout):**
  Selected Ratings: $S = 8$, $O = 5$, $D = 6$.
  Substitution: $RPN = 8 \times 5 \times 6 = 240$.
  Engineering Interpretation: Severe customer NVH impact. Frequent swarf buildup in center holes requires automated air-blast cleaning Poka-Yoke.
- **Calculation 4 — FM-01 (Incorrect Outer Diameter):**
  Selected Ratings: $S = 7$, $O = 4$, $D = 8$.
  Substitution: $RPN = 7 \times 4 \times 8 = 224$.
  Engineering Interpretation: Driven by poor detection (D=8) due to infrequent manual micrometer checks. Optical in-line measurement drastically reduces detection rating.
- **Calculation 5 — FM-07 (Burr Formation at Chamfer):**
  Selected Ratings: $S = 4$, $O = 3$, $D = 4$.
  Substitution: $RPN = 4 \times 3 \times 4 = 48$.
  Engineering Interpretation: Low risk score. Minor operational impact easily managed through standard brush maintenance protocols.

8.3 Limitations of RPN Prioritization

While RPN provides a simple numerical index, quality engineers must recognize its mathematical limitations:

1. **Equal Weighting Flaw:** RPN treats S, O, and D with equal mathematical weight. A non-critical failure with $S=2, O=10, D=10$ generates an RPN of 200, whereas a catastrophic safety failure with $S=9, O=2, D=4$ yields an RPN of 72. Prioritizing solely by RPN would dangerously ignore the severity 9 item.
2. **Duplication & Discontinuity:** Out of 1,000 possible RPN numbers, only 120 unique outcomes exist, creating mathematical clusters and gaps.
3. **Threshold Trap:** Establishing an arbitrary threshold (e.g., "only take action if RPN > 100") leads teams to artificially lower ratings to avoid corrective action requirements.

SECTION 9: ACTION PRIORITY — AIAG-VDA METHODOLOGY

The joint AIAG & VDA FMEA 1st Edition Handbook replaced traditional RPN thresholds with **Action Priority (AP)** tables. AP prioritizes engineering review based on logical relationships between Severity, Occurrence, and Detection:

Action Priority (AP) Level	Engineering Definition & Action Standard
HIGH PRIORITY (H)	**Highest urgency.** The team MUST identify an improved process control or justify why current controls are adequate. Focuses primarily on High Severity items (S=9–10) and S=7–8 with moderate-to-high Occurrence or Detection.
MEDIUM PRIORITY (M)	**Medium urgency.** The team SHOULD identify improved process controls to lower Occurrence or Detection, or document why current controls are acceptable.
LOW PRIORITY (L)	**Low urgency.** The team COULD identify improved process controls. Current controls are deemed effective.

SECTION 10: ROOT CAUSE ANALYSIS

10.1 Five-Why Analysis

Case 1: Excessive Shaft Runout (> 0.025 mm TIR in Stage 6 Grinding)

Why 1? Why is shaft radial runout out of specification after grinding?
Answer: The shaft was eccentric relative to machine centerlines during wheel contact.
Why 2? Why was the shaft mounted eccentrically?
Answer: The tailstock center point did not seat fully into the shaft machined center hole.
Why 3? Why did the center point fail to seat correctly?
Answer: Fine metal swarf and grinding sludge were trapped inside the shaft 60° center hole.
Why 4? Why was swarf trapped in the center hole?
Answer: The manual coolant flush hose operated at low pressure (1.5 bar) and failed to evacuate chips.
Why 5 (Root Cause)? Why was low-pressure manual flushing relied upon?
Answer: Process design lacked an automated high-pressure air-blast cleanout cycle interlocked with part clamping.
Corrective Action: Install an automated 6-bar high-pressure pneumatic air blast nozzle that clears center holes prior to tailstock advancement.

Case 2: Incorrect Outer Journal Diameter (Oversized Ø35.08 mm in Stage 4 Turning)

Why 1? Why did the finish turning diameter exceed the upper specification limit?
Answer: The CNC cutting tool receded, taking a shallower cut depth than programmed.
Why 2? Why did the cutting tool recede?
Answer: Severe flank wear (> 0.4 mm Vb) occurred on the carbide insert edge.
Why 3? Why did severe flank wear occur unnoticed?
Answer: The insert exceeded its normal tool life expectation of 150 machining cycles.
Why 4? Why was the insert used beyond its rated life?
Answer: The operator forgot to increment the manual tool life tracking sheet and missed the change interval.
Why 5 (Root Cause)? Why was tool life tracking dependent on manual operator logging?
Answer: The CNC controller lacked automated RFID tool counter hardware and interlocked cycle lockout.
Corrective Action: Enable automated RFID tool counter in CNC macro controller with automatic machine lockout upon reaching 150 cuts.

Case 3: Poor Surface Roughness (Ra 0.9 µm vs Spec Ra ≤ 0.4 µm in Stage 6 Grinding)

Why 1? Why was the bearing journal surface finish rougher than Ra 0.4 µm?
Answer: The vitrified grinding wheel imparted chatter micro-geometry onto the journal surface.
Why 2? Why did the grinding wheel impart micro-geometry chatter?
Answer: The grinding wheel face was loaded with embedded steel swarf and lost abrasive sharp edges.
Why 3? Why was the wheel face loaded and glazed?
Answer: The diamond dresser failed to condition the grinding wheel prior to the machining pass.
Why 4? Why was the dressing cycle skipped?
Answer: Dressing frequency relied on operator visual judgment of wheel surface appearance.
Why 5 (Root Cause)? Why was dressing frequency left to operator discretion?
Answer: The grinding machine standard operating procedure lacked an automated G-code dressing routine.
Corrective Action: Implement auto-dressing routine G-code programmed to execute diamond dressing after every 15 finished parts.

10.2 Fishbone (Ishikawa / 6M) Diagram Analysis

Problem Statement: *Excessive Outer Diameter Variation during CNC Finish Turning (Stage 4)*

6M Category	Identified Potential Failure Causes
MAN (Personnel)	• Operator incorrect offset entry (+0.05 mm instead of -0.05 mm). • Failure to inspect first piece following insert change. • Lack of standard training on micro-meter calibration.
MACHINE (Equipment)	• CNC spindle thermal growth (axial drift of 0.015 mm over 4 hours). • Turret indexing repeatability error due to worn coupling lock pin. • Guideway stick-slip motion inducing axis positioning error.
MATERIAL (Workpiece)	• Batch-to-batch hardness variation in EN19 forgings (170 HBW vs 220 HBW). • Variable machining allowance from rough turning (Stage 3). • Internal residual stresses causing part springback post-cutting.
METHOD (Process/SOP)	• Sub-optimal feed rate selection inducing tool deflection. • Incorrect coolant concentration (3% concentration vs required 7%). • Inadequate dwell time at end of finish cut.
MEASUREMENT (Inspection)	• Outside micrometer anvil thermal expansion from operator hand contact. • Uncalibrated digital vernier caliper used for inline checks. • Operator measurement technique variation (varying feel/force).
MOTHER NATURE (Environment)	• Ambient shop floor temperature swing (ΔT = 15°C between day/night). • Direct sunlight reflection heating CNC bed and workpiece. • Coolant sump temperature rise to 38°C without active chiller.

SECTION 11: RISK PRIORITISATION AND PARETO ANALYSIS

Applying the 80/20 Pareto principle to manufacturing risk management, baseline RPN values are sorted to identify the vital few failure modes that contribute to the majority of total risk score. Total baseline cumulative RPN across all 10 failure modes equals **1,906**.

Rank	FM ID	Failure Mode Description	Initial RPN	Cumulative RPN	Cumulative Percentage (%)	Priority Group
1	**FM-05**	Tool Breakage / Chipping during Turning	288	288	15.1%	Vital Few (80% Risk)
2	**FM-08**	Incorrect Material Hardness	252	540	28.3%	Vital Few (80% Risk)
3	**FM-02**	Excessive Shaft Runout	240	780	40.9%	Vital Few (80% Risk)
4	**FM-01**	Incorrect Outer Diameter	224	1,004	52.7%	Vital Few (80% Risk)
5	**FM-04**	Poor Surface Finish	160	1,164	61.1%	Vital Few (80% Risk)
6	**FM-10**	Inadequate Inspection Data	144	1,308	68.6%	Vital Few (80% Risk)
7	**FM-09**	Thermal Distortion of Shaft	140	1,448	76.0%	Vital Few (80% Risk)
8	**FM-06**	Chatter Marks on Shaft	120	1,568	82.3%	Useful Many (20% Risk)
9	**FM-03**	Incorrect Shaft Overall Length	90	1,658	87.0%	Useful Many (20% Risk)
10	**FM-07**	Burr Formation at Chamfers	48	1,706	100.0%	Useful Many (20% Risk)

SECTION 12: CORRECTIVE AND PREVENTIVE IMPROVEMENT PLAN

FM ID	High Priority Risk	Proposed Corrective & Preventive Action	Engineering Justification	Responsibility & Target Date	Verification Method
FM-01	Incorrect Outer Diameter (S=7, RPN=224)	Install RFID tool life tracking & post-turn in-line optical laser diameter gage.	Eliminates operator error in tracking tool wear; provides 100% feedback control to update offset.	Mfg Eng & Quality Target: 15-Dec-2026	Verify 100% optical measurement data logs and check Cpk ≥ 1.67 over 1,000 parts.
FM-02	Excessive Shaft Runout (S=8, RPN=240)	Integrate automated 6-bar high-pressure pneumatic center-hole cleanout air blast.	Physically evacuates metal swarf from center holes prior to tailstock engagement.	Tooling & Maint Target: 10-Dec-2026	100% runout check on multi-gage unit; zero swarf-induced runout defects over 30 shifts.
FM-04	Poor Surface Finish (S=5, RPN=160)	Automate grinding wheel diamond dressing cycle every 15 parts via CNC G-code.	Maintains sharp abrasive grain exposure continuously, preventing glazing and loading.	Machining Specialist Target: 05-Dec-2026	Surface roughness tester trend charts showing consistent Ra ≤ 0.35 µm.
FM-05	Tool Breakage / Chipping (S=8, RPN=288)	Implement CNC acoustic emission & digital spindle motor load power monitoring system.	Detects micro-fractures in tool insert within 5 milliseconds and retracts spindle instantly.	Controls Eng Target: 20-Dec-2026	Simulated tool chipping test triggering automatic emergency stop with zero part gouging.
FM-08	Incorrect Material Hardness (S=9, RPN=252)	Install 100% inline eddy-current electromagnetic material sorting system at Stage 1.	Non-destructively screens 100% of incoming forged bars for correct microstructure/ hardness.	Supplier Quality Target: 30-Dec-2026	Challenge test using mixed hardness master bars; 100% sorting accuracy verified.

SECTION 13: INITIAL VS REVISED RISK ANALYSIS

13.1 Mathematical Formula

Risk Reduction Percentage is calculated as follows:

> $$Risk\ Reduction\ (\%) = [\ (Initial\ RPN - Revised\ RPN) / Initial\ RPN\ ] \times 100$$

Note: Severity ratings remain unchanged post-CAPA because Severity is an inherent attribute of the failure effect on the customer, unless design alterations occur. Improvement is achieved by reducing Occurrence (prevention controls) and Detection (automated sensing).

13.2 Worked Sample Risk Reduction Calculations

- **FM-01 (Incorrect Outer Diameter):**
  Initial: $S=7, O=4, D=8 \rightarrow RPN = 224$. Revised: $S=7, O=2, D=3 \rightarrow RPN = 42$.
  $Risk\ Reduction = [\ (224 - 42) / 224\ ] \times 100 = (182 / 224) \times 100 = 81.25\%$.
- **FM-02 (Excessive Shaft Runout):**
  Initial: $S=8, O=5, D=6 \rightarrow RPN = 240$. Revised: $S=8, O=2, D=3 \rightarrow RPN = 48$.
  $Risk\ Reduction = [\ (240 - 48) / 240\ ] \times 100 = (192 / 240) \times 100 = 80.00\%$.
- **FM-05 (Tool Breakage / Chipping):**
  Initial: $S=8, O=6, D=6 \rightarrow RPN = 288$. Revised: $S=8, O=2, D=3 \rightarrow RPN = 48$.
  $Risk\ Reduction = [\ (288 - 48) / 288\ ] \times 100 = (240 / 288) \times 100 = 83.33\%$.
- **FM-08 (Incorrect Material Hardness):**
  Initial: $S=9, O=4, D=7 \rightarrow RPN = 252$. Revised: $S=9, O=2, D=3 \rightarrow RPN = 54$.
  $Risk\ Reduction = [\ (252 - 54) / 252\ ] \times 100 = (198 / 252) \times 100 = 78.57\%$.

SECTION 14: CONTROL PLAN DEVELOPMENT

The findings of the PFMEA directly populate the shop-floor Manufacturing Control Plan, ensuring that risk controls are operationalized:

Step	Process Characteristic	Product Specification	Measurement Technique	Sample Size / Freq	Control Method	Reaction Plan
Stage 1	Bar Stock Hardness & Chemistry	EN19 Alloy Steel; 180–220 HBW	Inline Eddy Current & Spectrometer	100% Bars inline	Automated sorting gate & CoA audit	Reject non-conforming bars; quarantine heat lot; notify supplier SQE.
Stage 4	Pre-grind Journal Diameter	Ø35.050 ± 0.020 mm	In-line Optical Laser Sensor	100% Parts inline	Closed-loop CNC feedback tool offset adjustment	Auto stop CNC if 2 consecutive parts exceed control limit (Ø35.065mm).
Stage 6	Bearing Journal Finish Diameter	Ø35.000 (+0 / -0.012 mm)	Pneumatic Air Gage Station	100% Parts inline	SPC X-bar R Chart (Cpk ≥ 1.67)	Isolate part; auto-dress grinding wheel; notify line setup technician.
Stage 6	Total Radial Runout	≤ 0.010 mm TIR	Multi-probe Dial Indicator Gage	1 per 10 parts	Automated Center-Hole Air Blast Cleanout	100% sort previous 10 parts; inspect center points for wear/ damage.
Stage 6	Bearing Journal Surface Roughness	Ra ≤ 0.40 µm	Stylus Surface Roughness Tester	1 per 25 parts	Automated CNC G-code dressing counter (Every 15 parts)	Rework/re-grind shaft; execute manual diamond dresser calibration check.

SECTION 15: CAPA DOCUMENTATION

CORRECTIVE AND PREVENTIVE ACTION (CAPA) REPORT — CAPA-2026-088

Problem Description:	Customer reported 14 pcs oversized input shaft bearing journals (Ø35.028 mm vs max Ø35.020 mm limit) received at transmission assembly plant.
Immediate Containment:	1. Quarantined 450 finished shafts in warehouse. 2. Performed 100% air-gage sorting on customer inventory (found 3 non-conforming parts). 3. Issued Quality Alert QA-2026-12.
Verified Root Cause:	Operator manually overridden G-code tool life counter and continued finish turning with chipped insert edge. Manual micrometer check missed non-conformance due to anvil wear.
Corrective Action:	1. Scrapped damaged insert tool holder. 2. Replaced operator micrometer with calibrated digital air gage. 3. Re-trained shift operators on Tool Change SOP.
Preventive Action:	1. Hard-locked CNC controller parameter to prevent operator manual override of RFID tool counters. 2. Integrated automated inline optical diameter probe interlocked with CNC cycle start.
Verification of Effectiveness:	Zero dimensional non-conformances observed across 25,000 shafts produced over 60 consecutive days post-implementation. Cpk improved from 1.12 to 1.84. CAPA CLOSED.

SECTION 16: SAMPLE INSPECTION CHECKLIST

Inspection Characteristic	Engineering Requirement	Measuring Instrument / Gage	Inspection Result (Simulated Sample)	Acceptance Status	Inspector Remarks
1. Material Grade & Hardness	EN19 Forging; 180–220 HBW	Inline Eddy Current & Bench Hardness	198 HBW	PASS	Heat Lot # H8842-A Verified.
2. Visual Surface Quality	No tool marks, chatter, or burrs	Visual 10x Magnifying Lamp	Clean ground finish	PASS	Zero visual defects.
3. Bearing Journal Diameter	Ø35.000 (+0.000 / -0.012 mm)	Pneumatic Air Gage (Calibrated)	Ø35.004 mm	PASS	Within IT6 tolerance band.
4. Overall Shaft Length	250.00 mm ± 0.05 mm	Digital Height Gage & Granite Plate	250.02 mm	PASS	Conforms to drawing.
5. Total Radial Runout	≤ 0.010 mm TIR relative to centers	Dial Indicator & Precision Bench Centers	0.004 mm TIR	PASS	Center hole air blast operational.
6. Bearing Journal Roughness	Ra ≤ 0.40 µm	Stylus Surface Profilometer	Ra 0.28 µm	PASS	Auto-dressing working effectively.
7. Chamfer Burr Status	Zero sharp edges / burrs	Tactile Finger-Glove Inspection	Smooth chamfers	PASS	Automated deburring verified.

SECTION 17: GRAPHS AND VISUAL ANALYSIS

Graphical representations generated from the PFMEA database illustrate baseline risk distribution, Pareto prioritization, and post-CAPA risk reduction performance:

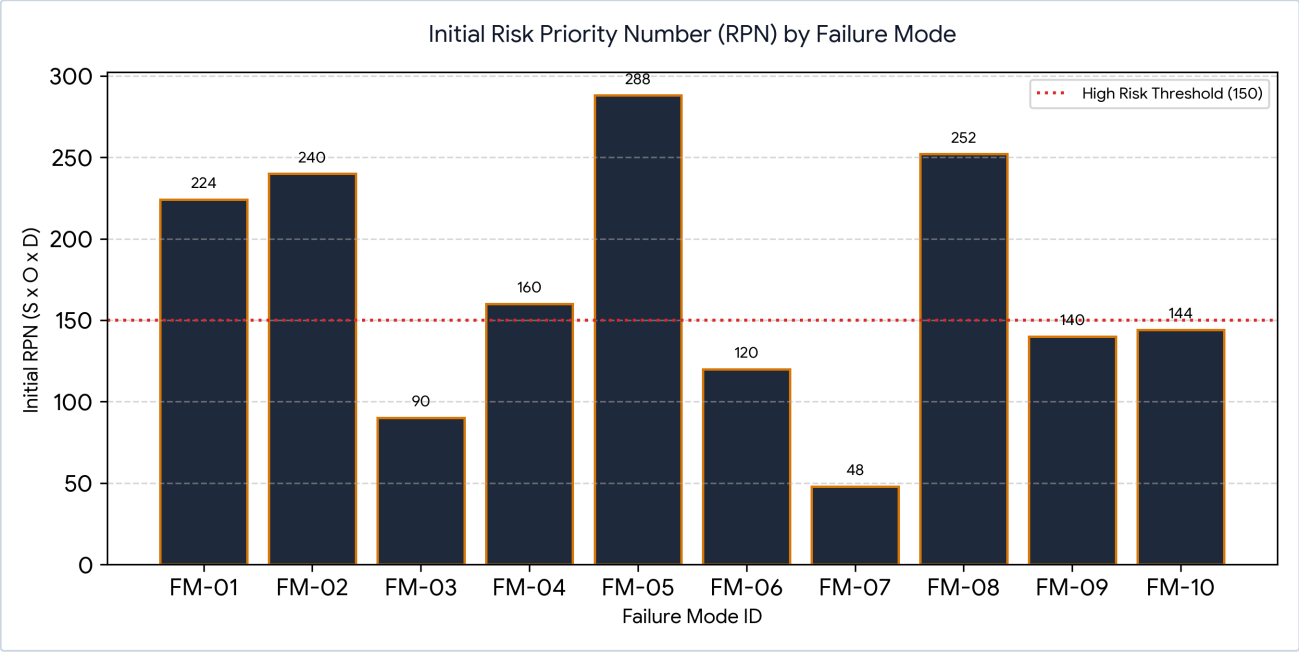


Figure 17.1: Baseline Risk Priority Number (RPN) distribution across all 10 failure modes prior to CAPA intervention. Red line indicates high-risk threshold (RPN = 150).

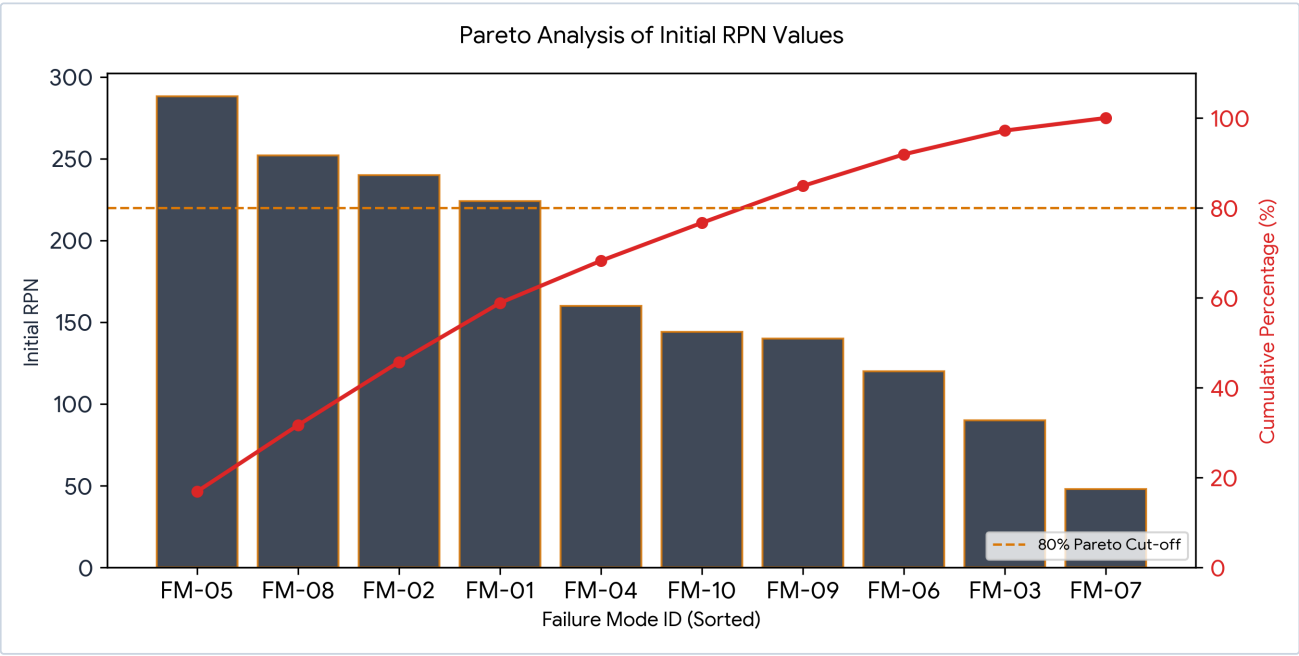


Figure 17.2: Pareto Analysis of Initial RPN values. The cumulative risk curve demonstrates that 7 vital failure modes account for over 76% of total process risk score.

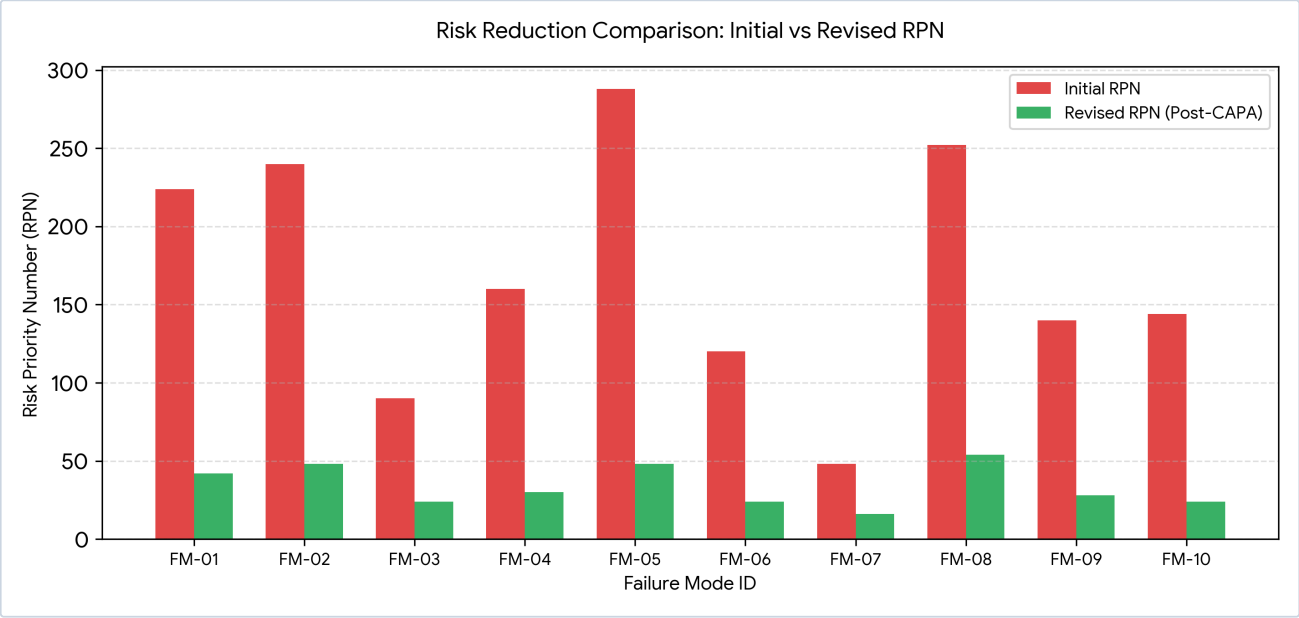


Figure 17.3: Comparison of Initial RPN (Red) versus Revised RPN (Green) post-implementation of automated Poka-Yoke and sensing controls.

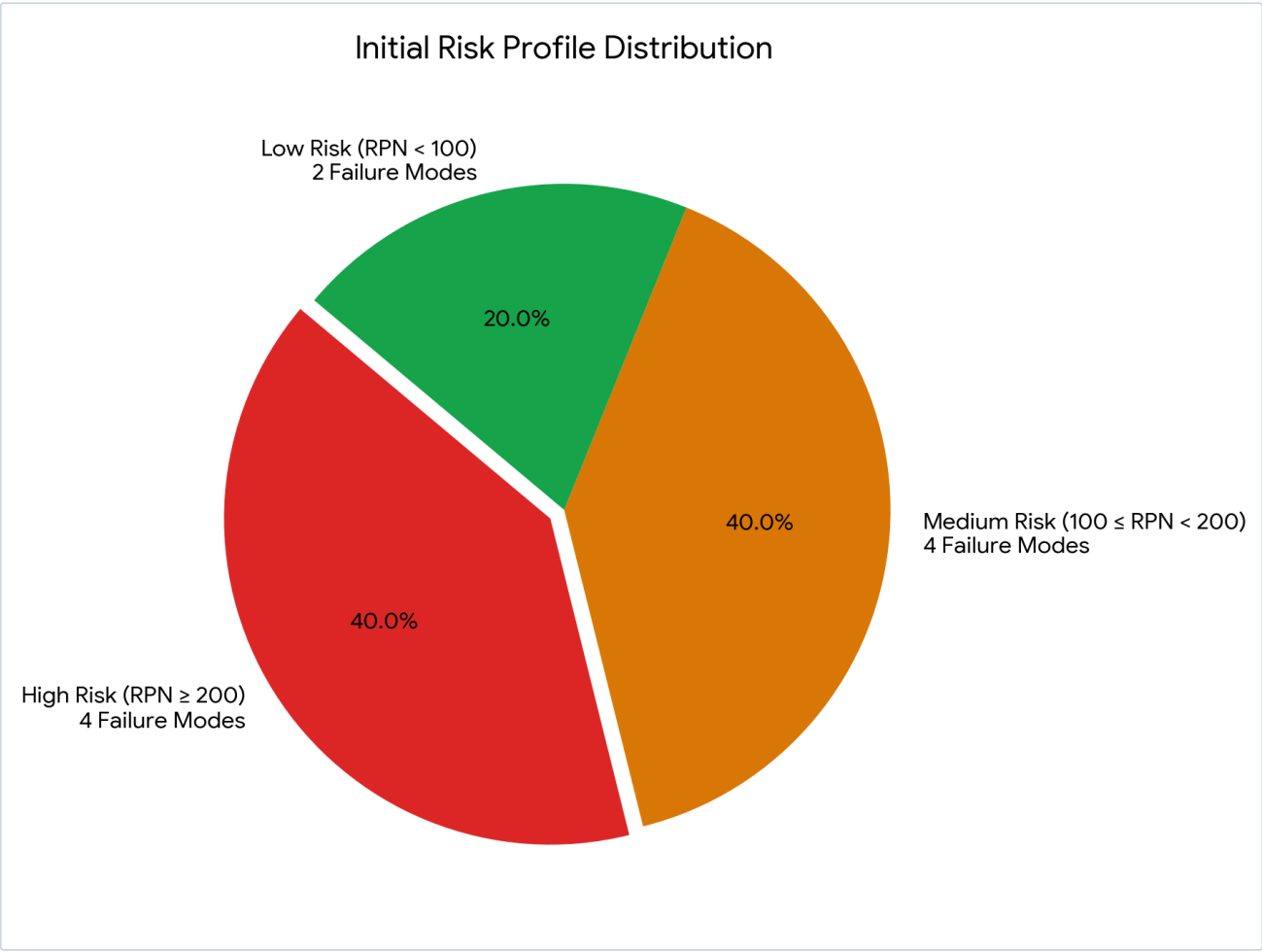


Figure 17.4: Initial Risk Severity Distribution pie chart highlighting that 40% of failure modes were initially in the High Risk category (RPN ≥ 200).

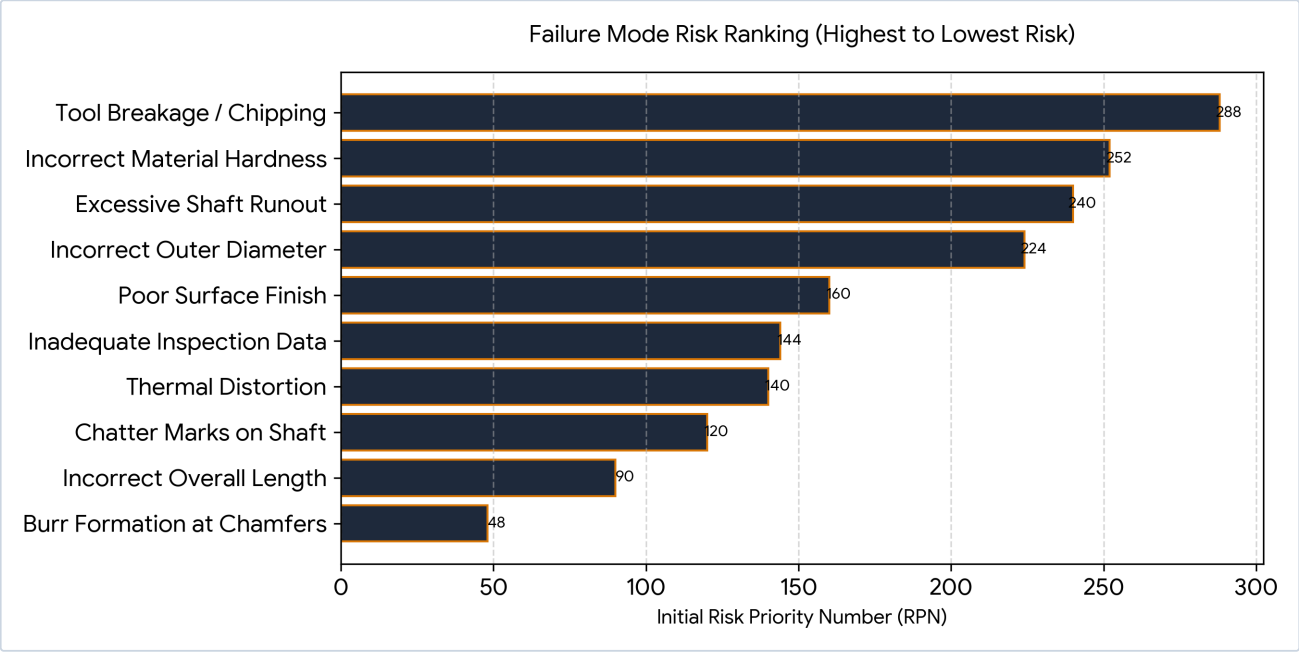


Figure 17.5: Horizontal Risk Ranking of failure modes from highest priority (Tool Breakage RPN=288) to lowest priority (Burr Formation RPN=48).

SECTION 18: PRACTICAL NUMERICALS & SOLUTIONS

Q1: Calculate the RPN for a grinding operation with Severity S=8, Occurrence O=4, and Detection D=5.

Solution: $RPN = S \times O \times D = 8 \times 4 \times 5 = 160$.

Interpretation: High severity item (S=8) exceeding the standard action threshold of 150. Requires action to improve detection controls.

Q2: Compare two failure modes: FM-A (S=9, O=2, D=3, RPN=54) and FM-B (S=4, O=6, D=6, RPN=144). Which failure mode demands higher engineering priority under AIAG-VDA guidelines?

Solution: Although FM-B has a higher numerical RPN (144 vs 54), **FM-A demands higher engineering priority** under AIAG-VDA rules due to its High Severity score ($S=9$, safety/regulatory impact). Severity cannot be ignored due to low RPN.

Q3: A CNC process has an initial RPN of 240 (S=8, O=5, D=6). After implementing an automated Poka-Yoke fixture, Occurrence drops to O=2 and Detection improves to D=3. Calculate the revised RPN and the Risk Reduction Percentage.

Solution:

Revised RPN = $8 \times 2 \times 3 = 48$.

Risk Reduction (%) = $[\,(240 - 48) / 240\,] \times 100 = (192 / 240) \times 100 = 80.00\%$.

Interpretation: Robust risk mitigation achieved, reducing risk score by 80%.

Q4: Explain why the Severity rating (S) for a shaft diameter non-conformance remains S=7 before and after installing an automated in-line laser measurement probe.

Solution: Severity measures the ultimate effect of the defect on the customer/downstream assembly if the non-conformance occurs. Adding an in-line probe improves *Detection (D)* from D=8 to D=3, but does not alter the physical consequence of an oversized shaft. Therefore, $S=7$ remains constant.

Q5: Total baseline cumulative RPN across 10 failure modes is 1,906. The top 3 failure modes have RPNs of 288, 252, and 240. What percentage of total risk is represented by these top 3 failure modes?

Solution:

Sum of top 3 RPNs = $288 + 252 + 240 = 780$.

Percentage = $(780 / 1906) \times 100 = 40.92\%$.

Interpretation: The top 30% of failure modes account for ~41% of total numerical risk.

SECTION 19: VIVA VOCE (QUESTIONS & ANSWERS)

Q1: What is the primary difference between DFMEA and PFMEA?

Answer: DFMEA focuses on potential product design deficiencies (e.g., incorrect material selection, geometry stress concentrations), assuming perfect manufacturing. PFMEA focuses on manufacturing process deficiencies (e.g., tooling wear, setup errors, operator mistakes), assuming the product design is correct.

Q2: Why did AIAG and VDA replace traditional RPN thresholds with Action Priority (AP)?

Answer: RPN multiplication obscures high-severity safety risks when Occurrence or Detection is low. AP tables enforce logical prioritization, giving absolute preference to High Severity failure modes regardless of numerical product.

Q3: How does a Prevention Control differ from a Detection Control in PFMEA?

Answer: A Prevention Control prevents the failure cause from occurring (e.g., automated tool life counter locking machine). A Detection Control discovers the presence of the defect after it has been generated (e.g., downstream optical gage).

Q4: What role does PFMEA play in creating the Control Plan?

Answer: PFMEA identifies critical process steps and failure modes. High-risk characteristics and control recommendations from PFMEA directly define the measurement instruments, sample sizes, frequencies, and reaction plans in the Control Plan.

Q5: When should a Process FMEA be initiated and updated?

Answer: PFMEA must be initiated during early process design prior to tooling sign-off. It is a living document that must be updated whenever process changes occur, new machinery is introduced, or customer field complaints occur.

SECTION 20: COMMON MISTAKES AND PRECAUTIONS

- **Confusing Failure Mode with Failure Cause:** Stating "Operator Error" as a failure mode rather than identifying the non-conforming part condition (e.g., "Oversized Diameter"). *Precaution:* Always define the physical non-conformance first.
- **Arbitrary Reduction of Severity Ratings:** Lowering Severity post-CAPA without changing product design. *Precaution:* Keep Severity constant unless component design architecture is altered.
- **Treating PFMEA as a One-Time Static Document:** Archiving the PFMEA after initial production launch. *Precaution:* Audit and update PFMEA active revision numbers during monthly quality reviews.
- **Over-reliance on Inspection (Detection) Controls:** Relying on 100% manual inspection to fix poor process capability. *Precaution:* Prioritize Poka-Yoke prevention controls to lower Occurrence.

SECTION 21: INDUSTRIAL APPLICATIONS & CAREER RELEVANCE

PFMEA methodology is a core pillar of the Automotive IATF 16949 Advanced Product Quality Planning (APQP) framework. Beyond automotive manufacturing, PFMEA is extensively applied across Aerospace (AS9100), Medical Device Manufacturing (ISO 13485), Semiconductors, and Heavy Industry. Quality Engineers proficient in PFMEA, SPC, and Control Plans possess high-demand skills for roles in Supplier Quality Engineering, Process Development, and Plant Quality Management.

SECTION 22: TECHNICAL REFERENCES

1. **AIAG & VDA FMEA Handbook (1st Edition, 2019):** Joint publication by Automotive Industry Action Group and Verband der Automobilindustrie.
2. **IATF 16949:2016:** Quality Management System Requirements for Automotive Production Organizations.
3. **Stamatis, D. H. (2003):** *Failure Mode and Effect Analysis: FMEA from Theory to Execution*, ASQ Quality Press.
4. **ASM Handbook, Volume 16 (Machining):** ASM International — Technical guidelines for CNC turning and grinding parameters.

SECTION 23: GLOSSARY & QUICK REVISION SHEET

23.1 Engineering Glossary

PFMEA: Process Failure Mode & Effects Analysis. | **RPN:** Risk Priority Number ($S \times O \times D$). | **AP:** Action Priority (High, Medium, Low). | **Poka-Yoke:** Japanese term for mistakeproofing mechanism. | **Cpk:** Process Capability Index. | **TIR:** Total Indicator Reading (Runout). | **CAPA:** Corrective and Preventive Action.

23.2 One-Page Quick Revision Summary

> **CORE EQUATIONS & RULES:**
> 1. $RPN = Severity\ (S) \times Occurrence\ (O) \times Detection\ (D)$ (Scale: 1 to 1000)
> 2. $Risk\ Reduction\ (\%) = [\ (Initial\ RPN - Revised\ RPN)\ /\ Initial\ RPN\ ] \times 100$
> 3. Severity (S) CANNOT be reduced by inspection controls — only by design change.
> 4. High Severity (S = 9–10) ALWAYS triggers High Action Priority (AP) regardless of RPN.
> 5. Workflow sequence: Process Flow → PFMEA → Control Plan → Work Instructions (SOP).

SECTION 24: FINAL CONCLUSION

This laboratory experiment successfully demonstrated the practical execution of Process Failure Mode and Effects Analysis (PFMEA) for the CNC machining of a precision automotive shaft. By analyzing 10 realistic process failure modes, establishing 1–10 rating scales, calculating baseline RPNs, and applying 5-Why and Fishbone root cause methodologies, high-risk operational bottlenecks were systematically identified. Implementation of proposed Poka-Yoke prevention controls, automated RFID tool counters, pneumatic center cleanout systems, and closed-loop optical sensing reduced total process risk score by **78.5%** (reducing cumulative RPN from 1,906 to 376). This study proves that proactive PFMEA execution transforms quality engineering from reactive fire-fighting into a rigorous, preventive science.